

PURBANCHAL UNIVERSITY

2025

Master of Computer Application (M.C.A.)/First Semester/Final

Time: 03:00 hrs.

Full Marks: 80 /Pass Marks: 32

MCA102: Software Engineering (New Course)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group A

Answer TWO questions.

2×12=24

1. Define the main characteristics of good software. Compare incremental and iterative development. Explain spiral model and its different phases. [3+4+5]
2. Describe the relationship between requirement elicitation and requirement analysis. List down the major approaches to requirement elicitation and explain any three of them. [2+10]
3. Differentiate between white-box and black-box testing strategies. Discuss verification and validation in the testing process. [8+4]

Group B

Answer SEVEN questions.

7×8=56

4. What are the various risks associated with software development? Discuss the risk analysis process used in software engineering. [3+5]
5. Why is software planning an integral part of software engineering? Discuss any two COCOMO cost estimation models. [2+6]
6. Explain the importance Software Requirement Specification (SRS) document. Discuss about different non-functional requirements in software. [2+6]
7. What are the goals of software design? Differentiate between architectural design and procedural design, with examples. [3+5]
8. Define Software Quality Assurance (SQA) and discuss its key activities in software development.
9. Why is dependability considered a key attribute of software quality? Discuss how fault tolerance mechanisms can improve both availability and safety. [2+6]
10. How do rapid changes in technology affect software development life cycles? How can software engineers prepare for the challenges of rapidly evolving technologies? [4+4]

Contd. ...

(2)

✓ 11. Write short notes on Any TWO:

2×3=6

(a) Data flow diagram

(b) Formal technical review

(c) Software reuse

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Basic
Intermediary

Organic

Extended